

MIDTERM II—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions. Please check these solutions. In order to have grounds to appeal for partial credit, your solutions should resemble what I write here in terms of stating properly the problem, with all its constraints, as well as making some progress in reaching a correct solution at the minimum.

I will solve the problem here in its general parametric form, since all of you got random values for the parameters.

Procurement Contracts—Problem 2 in the exam

A government offers a firm a military contractor to buy $q \geq 0$ units of military hardware, in exchange for a payment of $t \geq 0$. The contractor's constant unit cost is private information to it and is either high (c_h) or low (c_ℓ) with $c_h > c_\ell$. The government's beliefs about the military contractor's cost is given by $\Pr(c_\ell) = p \in (0, 1)$. If the firm with cost c_i , $i = h, \ell$, accepts the contract (t_i, q_i) , its profits is $t_i - c_i q_i$, whereas if it rejects it receives zero profit. The payoff for the government is given by

$$\ln(q) - t.$$

1. Assume that the government observes the firm's type. Find the First Best Contract.
2. Now assume that the government cannot observe the firm's type. Find the optimal contract. Compare with the first best.
3. Assume that now the government faces a contractor with *three possible* types: high, medium and low ($c_h > c_m > c_\ell$). The beliefs of the government regarding these costs are: $\Pr(c_\ell) = p_\ell \in (0, 1)$, $\Pr(c_m) = p_m \in (0, 1)$ and $\Pr(c_h) = p_h \in (0, 1)$. Characterize as much as you can the optimal contract in this case.

Solution to the problem

These are suggested solutions. Note that in here I prove formally which conditions bind and which do not, which is a step that you may have skipped and just stated in your own solutions.

1. The government's problems are, for each $i = \ell, h$

$$\begin{aligned} & \max_{(q_i, t_i)} \ln(q_i) - t_i \\ \text{s.t. } & t_i - c_i q_i \geq 0 \\ & q_i \geq 0. \end{aligned}$$

At the solution, the constraints must hold with equality, i.e.

$$t_i = c_i q_i.$$

Substituting into the objective function, the problem becomes:

$$\max_{q_i \geq 0} \ln(q_i) - c_i q_i$$

At the solution, we must have $q_i > 0$ since

$$\lim_{q_i \rightarrow 0} \ln(q_i) - c_i q_i = -\infty, \quad i = \ell, h.$$

Therefore, the following first order conditions must hold:

$$\frac{1}{q_i} = c_i, \quad i = \ell, h.$$

We can solve them as

$$q_i = \frac{1}{c_i}, \quad i = \ell, h.$$

Transfers are therefore

$$t_i = 1, \quad i = \ell, h.$$

2. The government's problem is now

$$\begin{aligned} \max_{(q_i, t_i)_{i=\ell, h}} \quad & p(\ln(q_\ell) - t_\ell) + (1-p)(\ln(q_h) - t_h) \\ \text{s.t.} \quad & t_\ell - c_\ell q_\ell \geq 0 \\ & t_h - c_h q_h \geq 0 \\ & t_\ell - c_\ell q_\ell \geq t_h - c_\ell q_h \\ & t_h - c_h q_h \geq t_\ell - c_h q_\ell \\ & q_\ell \geq 0, q_h \geq 0 \end{aligned}$$

The following must be true at the solution:

(a) Participation constraint of the high cost holds with equality, i.e. $t_h - c_h q_h = 0$.

Proof. Suppose, towards a contradiction, that $t_h - c_h q_h > 0$. Then, the incentive compatibility constraint of the low cost implies that

$$t_\ell - c_\ell q_\ell \geq t_h - c_\ell q_h > t_h - c_h q_h > 0$$

This cannot be a solution since the government can decrease both t_ℓ and t_h by the same small amount without violating any of the constraints and increase its payoff. $\square$

(b) Participation constraint of the low cost does not bind at the solution.

Proof. IC constraint of the low cost and the PC constraint of the high cost imply that

$$t_\ell - c_\ell q_\ell \geq t_h - c_\ell q_h > t_h - c_h q_h \geq 0.$$

$\square$

(c) Incentive compatibility of the low cost holds with equality, i.e. $t_\ell - c_\ell q_\ell = t_h - c_\ell q_h$.

Proof. Suppose, towards a contradiction, that $t_\ell - c_\ell q_\ell > t_h - c_\ell q_h$. Then,

$$t_\ell - c_\ell q_\ell > t_h - c_\ell q_h > t_h - c_h q_h \geq 0$$

Therefore, the government can decrease t_ℓ by some small amount without violating any of the constraints and increase its payoff. $\square$

(d) $q_\ell \geq q_h$.

Proof. The two IC constraints can be written as

$$c_\ell(q_\ell - q_h) \leq t_\ell - t_h \leq c_h(q_\ell - q_h)$$

Since $c_\ell < c_h$, we must have that $q_\ell \geq q_h$. $\square$

(e) Incentive compatibility of the high cost can be ignored.

Proof. Since the IC constraint for the low cost holds with equality and $q_\ell \geq q_h$, we have

$$t_\ell - t_h = c_\ell(q_\ell - q_h) \leq c_h(q_\ell - q_h)$$

which implies

$$t_h - c_h q_h \geq t_\ell - c_h q_\ell$$

□

Therefore, the government's problem is equivalent to the following relaxed problem

$$\begin{aligned} \max_{(q_i, t_i)_{i=\ell, h}} \quad & p(\ln(q_\ell) - t_\ell) + (1-p)(\ln(q_h) - t_h) \\ \text{s.t.} \quad & t_h - c_h q_h = 0 \\ & t_\ell - c_\ell q_\ell = t_h - c_\ell q_h \\ & q_\ell \geq q_h \\ & q_\ell \geq 0, q_h \geq 0 \end{aligned}$$

From the equality constraints we have

$$\begin{aligned} t_h &= c_h q_h \\ t_\ell &= c_\ell q_\ell + (c_h - c_\ell) q_h \end{aligned}$$

Substituting into the objective we obtain

$$\begin{aligned} \max_{q_\ell, q_h} \quad & p(\ln(q_\ell) - c_\ell q_\ell - (c_h - c_\ell) q_h) + (1-p)(\ln(q_h) - c_h q_h) \\ \text{s.t.} \quad & q_\ell \geq q_h \\ & q_\ell \geq 0, q_h \geq 0 \end{aligned}$$

Let us first ignore the monotonicity condition. At the solution, we must have $q_\ell > 0$ and $q_h > 0$, and hence, the following first order conditions must hold:

$$\begin{aligned} p \left(\frac{1}{q_\ell} - c_\ell \right) &= 0 \\ -p(c_h - c_\ell) + (1-p) \left(\frac{1}{q_h} - c_h \right) &= 0 \end{aligned}$$

The previous equations can be solved as

$$\begin{aligned} q_\ell &= \frac{1}{c_\ell} \\ q_h &= \frac{1}{c_h + \frac{p}{1-p}(c_h - c_\ell)} \end{aligned}$$

Since $c_\ell < c_h$, we have that $q_\ell > q_h$, i.e., the monotonicity condition is satisfied. Therefore, this is indeed the solution. Transfers can be calculated using

$$\begin{aligned} t_h &= c_h q_h \\ t_\ell &= c_\ell q_\ell + (c_h - c_\ell) q_h \end{aligned}$$

The low cost type produces the efficient (First-best) quantity and receives $(c_h - c_\ell)q_h$ as information rent, whereas the high cost produces less than the efficient amount and receives zero rent. High cost type's quantity is distorted to reduce the information rent of the low cost. The amount of distortion increases in the cost difference $c_h - c_\ell$, and in the fraction of low cost types p .

3. The monopolist problem is now

$$\begin{aligned}
& \max_{(q_i, t_i)_{i=\ell, m, h}} \sum_{i \in \{\ell, m, h\}} p_i(\ln(q_i) - t_i) \\
& \text{s.t.} \quad t_i - c_i q_i \geq 0, \quad i = \ell, m, h \\
& \quad t_i - c_i q_i \geq t_j - c_j q_j, \quad i, j \in \{\ell, m, h\} \\
& \quad q_i \geq 0, \quad i = \ell, m, h
\end{aligned}$$

Following problem set 3 and the lecture slides, the following properties, which I just state without proof, must be true at the solution:

- (a) The participation constraint of the high cost holds with equality, i.e. $t_h - c_h q_h = 0$.
- (b) All the other participation constraints do not bind at the solution.
- (c) The only incentive compatibility constraints that hold with equality are:

$$\begin{aligned}
t_\ell - c_\ell q_\ell &= t_m - c_\ell q_m \\
t_m - c_m q_m &= t_h - c_m q_h
\end{aligned}$$

- (d) All other incentive constraints do not bind at the solution.
- (e) $q_\ell \geq q_m \geq q_h$.

Thus, the government's problem is equivalent to the following relaxed problem:

$$\begin{aligned}
& \max_{(q_i, t_i)_{i=\ell, m, h}} \sum_{i \in \{\ell, m, h\}} p_i(\ln(q_i) - t_i) \\
& \text{s.t.} \quad t_h - c_h q_h = 0 \\
& \quad t_\ell - c_\ell q_\ell = t_m - c_\ell q_m \\
& \quad t_m - c_m q_m = t_h - c_m q_h \\
& \quad q_\ell \geq q_m \geq q_h \\
& \quad q_i \geq 0, \quad i = \ell, m, h
\end{aligned}$$

Ignoring the monotonicity condition, we set up the Lagrangian:

$$\begin{aligned}
\mathcal{L} = & \sum_{i \in \{\ell, m, h\}} p_i(\ln(q_i) - t_i) + \lambda_h(t_h - c_h q_h) + \lambda_m(t_m - t_h - c_m q_m + c_m q_h) \\
& + \lambda_\ell(t_\ell - t_m - c_\ell q_\ell + c_\ell q_m) + \sum_{i \in \{\ell, m, h\}} \mu_i q_i
\end{aligned}$$

The KT conditions for optimality are:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial q_i} &= 0 \quad i = \ell, m, h \\
\frac{\partial \mathcal{L}}{\partial t_i} &= 0 \quad i = \ell, m, h \\
\frac{\partial \mathcal{L}}{\partial \lambda_i} &= 0 \quad i = \ell, m, h \\
\mu_i q_i &= 0, \mu_i \geq 0, q_i \geq 0 \quad i = \ell, m, h
\end{aligned}$$

For the moment, we will focus on solutions such that $q_i > 0$, which implies that $\mu_i = 0$ for all i .

For $i = h$ we have

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial q_h} &= \frac{p_h}{q_h} - \lambda_h c_h + \lambda_m c_m = 0 \\ \frac{\partial \mathcal{L}}{\partial t_h} &= -p_h + \lambda_h - \lambda_m = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_h} &= t_h - c_h q_h = 0\end{aligned}$$

For $i = m$ we have

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial q_m} &= \frac{p_m}{q_m} - \lambda_m c_m + \lambda_\ell c_\ell = 0 \\ \frac{\partial \mathcal{L}}{\partial t_m} &= -p_m + \lambda_m - \lambda_\ell = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_m} &= t_m - t_h - c_m q_m + c_m q_h = 0\end{aligned}$$

For $i = \ell$ we have

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial q_\ell} &= \frac{p_\ell}{q_\ell} - \lambda_\ell c_\ell = 0 \\ \frac{\partial \mathcal{L}}{\partial t_\ell} &= -p_\ell + \lambda_\ell = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_\ell} &= t_\ell - t_m - c_\ell q_\ell + c_\ell q_m = 0\end{aligned}$$

Thus we have that $\lambda_\ell = p_\ell$, $\lambda_m = p_m + p_\ell$ and that $\lambda_h = p_h + p_m + p_\ell$. From this we get that:

$$\begin{aligned}q_\ell &= \frac{1}{c_\ell} \\ q_m &= \frac{1}{c_m + \frac{p_\ell}{p_m}(c_m - c_\ell)} \\ q_h &= \frac{1}{c_h + \frac{p_m + p_\ell}{p_h}(c_h - c_m)}\end{aligned}$$

Notice that because $1/x > 0$ for all $x > 0$ and all costs and probabilities are positive in the problem, we will not have corner solutions. Thus we were right to assume initially that the solutions would be strictly positive.

Once again we have that there is no distortion at the top, i.e., q_ℓ is equal to the efficient quantity. For types with medium and high costs, there will be a distortion in quantity.

Finally, transfers can be obtained from the following equalities:

$$\begin{aligned}t_h &= c_h q_h \\ t_m &= c_m(q_m - q_h) + t_h \\ t_\ell &= c_\ell(q_\ell - q_m) + t_m\end{aligned}$$

Remark! Note that in the solution above $q_\ell > q_m$. However, it not obvious when we will have $q_m \geq q_h$. Remember that when solving the problem, we usually ignore the monotonicity restriction, and then we need to check that it holds in the end. For the general parametric problem, we can proceed in two ways:

(a) *Necessary and sufficient condition:*

$$\begin{aligned}
q_m \geq q_h &\Leftrightarrow \frac{1}{c_m + \frac{p_\ell}{p_m}(c_m - c_\ell)} \geq \frac{1}{c_h + \frac{p_m + p_\ell}{p_h}(c_h - c_m)} \\
&\Leftrightarrow c_m + \frac{p_\ell}{p_m}(c_m - c_\ell) \leq c_h + \frac{p_m + p_\ell}{p_h}(c_h - c_m) \\
&\Leftrightarrow \frac{p_\ell}{p_m}(c_m - c_\ell) \leq (c_h - c_m) \left(\frac{p_m + p_\ell}{p_h} \right) \\
&\Leftrightarrow \frac{\frac{p_\ell}{p_m}}{\frac{p_h + p_m + p_\ell}{p_h}} = \frac{p_h p_\ell}{p_m} \leq \frac{c_h - c_m}{c_m - c_\ell}
\end{aligned}$$

(b) *Sufficient conditions:* The following inequalities must hold

$$\frac{p_\ell}{p_m} \leq \frac{p_m + p_\ell}{p_h} \quad c_m - c_\ell \leq c_h - c_m$$

The sufficient conditions are the ones usually found in the literature: for this problem, where the ordering of costs is $c_h > c_m > c_\ell$, we need an increasing inverse hazard rate and that the difference in costs be increasing. (Check slide 38 in Lecture 6: Screening II—General Models.)

You did not need to know the above to solve your exam, because all your solutions were supposed to satisfy monotonicity. I made sure that this would happen by picking values for the random parameters that would satisfy the *sufficient conditions* described above.